

EVERY CAR GETS ITS OWN GARAGE. EVERY CARD STAYS ON SHOW.

PITLANE is a 3D-printed wall system for carded die-cast cars. One 40 cm strip holds up to 7 cars, and each card sits in its own swing-out ledge. Lift a ledge, swing it open, slide the card in, swing it shut. The cards step out 7 mm per row, so every name stays visible and no card covers another.

The pitch (30 seconds): "Collectors keep their best cars carded, then hide them in boxes or crush them on peg hooks. PITLANE gives every card its own swing-out garage on the wall. You can see every name, take any one card out without touching the others, and add racks as your collection grows. It snaps together with no tools. One 40 cm strip holds up to 7 cars, at about the price of a few mainline cars."



Real printed prototype loaded with real carded cars (designer's photo).



Current design: 7 cars on one 40 cm strip (CAD render, cards shown as blanks).

WHAT'S IN THE BOX (7-CAR KIT)

1x wall strip, 40 cm

One piece, or two halves that lock with a C-interlock. Fixed with 4 screws. Hook-lip pairs every 54 mm.

7x hinge clips

Each clip double-hooks behind two lips and clicks onto a friction bump. Each one sits 7 mm deeper than the one below. No tools.

7x swing ledges

8 mm pin running the full height, a 0° stop, a click detent and 14 mm corner posts that keep the card upright.

WHY IT WINS

- **Grab one, leave the rest:** other displays make you unhook the cards in front. Here each card has its own door.
- **Names stay readable:** a 1 mm front lip and stepped rows mean nothing covers the card title.
- **Built for real prints:** every part prints without supports, and every lock was redesigned after real test prints failed.
- **Grows with the collection:** buy the strip once and add racks for about \$6 each.

MARKET

Adult collectors now drive the die-cast market: an estimated 58% of revenue, worth about \$4.2–5.8 billion worldwide in 2025 and growing 5–6% a year. North America is the largest region at about 38%. Many collectors keep their cars **carded** (mint in package), which is the segment PITLANE serves.

COMPETITION AND PRICES (OCT 2026)

Product	What it does	Price	PITLANE advantage
Etsy 3D-printed hooks and card stands	Static hooks or stands; cards hang flat or in rows	\$2.75–\$12.86 (e.g. 20 hangers \$12.86, 4-pack \$12)	Swing-out access, rows that step forward, double-lock clip
Official collector “case strips” (Target)	5 clear sleeves, 10 cars each, stacked	\$9.99 for 50 cars	Each card visible and removable on its own; no plastic sleeves
Wall cabinets / display cases	Loose (uncarded) cars behind glass	\$36–\$124 (24–56 cars)	Made for <i>carded</i> cars; much cheaper per wall metre

Sources: [Etsy carded displays](#), [Etsy 20-hanger display](#), [Target collector case strips](#), [Walmart displays](#), [Market Intelto](#), [Intel Market Research](#).

SWOT

Strengths

- A genuinely new mechanism: swing-out door for each card
- Every name visible; rows step out 7 mm
- No supports, low cost per unit, modular
- Designer is a collector, with real-print testing

Weaknesses

- Fits standard mainline cards only (~105 × 165 mm, blister ≤ 42 mm)
- Print time per kit is high (estimate 12–18 h)
- Ledges above must be opened to load a lower card
- The newest version (friction bump, L-lock) is not yet test-printed

Opportunities

- Sell the STL files too: zero marginal cost
- Colour editions (orange track, black and white)
- Premium / long-card ledge, Matchbox and Mini GT variants
- Hobby-shop wall displays (B2B)

Threats

- Cheap static hooks at \$3–13
- Copycats on file marketplaces
- Trademark: never use “Hot Wheels” as your brand, only “fits Hot Wheels® cards”
- Shipping cost of a 40 cm part

WHO BUYS IT

- **Adult carded collectors** (25–45) with 20–200 cards and a wall to fill. This is the main ad audience.
- **Parents** buying a tidy display for a child’s collection (gift season: Nov–Dec).
- **Makers** who print their own and buy the STL files on Printables, MakerWorld and Cults3D.

COST TO MAKE (ESTIMATE)

Item	4-car kit	7-car kit	Basis
Filament (PLA/PETG at about \$20/kg)	\$3.40–4.00	\$5.40–6.20	From the STL files: 281 g / 449 g if fully solid; real prints about 60–70% of that
Power and printer wear	\$1.50	\$2.50	About \$0.15/h × 10–16 h of printing
4 screws, wall plugs, packaging	\$2.00	\$2.50	Box for a 40 cm strip (or ship the 2-piece strip)
Total cost	≈ \$7–8	≈ \$10–11	Excludes your time and shipping (charge shipping separately)

RECOMMENDED PRICES (USD)

STL files \$7.99 All parts and kits, print at home. Launch at \$5.99. Sell on Etsy digital, Cults3D and MakerWorld.	Starter (4 cars) \$29.99 40 cm strip + 4 clips + 4 ledges. Room to add 3 more.	BEST VALUE Full strip (7 cars) \$39.99 Fully loaded. \$5.71 per car. Launch at \$34.99.	Add-on rack \$6.99 One clip + one ledge. Bundle of 3 for \$17.99.
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Why these prices: static Etsy hooks sell for \$3–13 and glass cabinets for \$36–124. PITLANE sits between them as a *functional premium* product at about \$5–6 per car. At \$39.99 on Etsy (roughly \$4.40 in fees) the 7-car kit leaves about **\$24–25 margin** before your time. Don't go below \$29.99 for the full strip: it signals a cheap hook, which this isn't.

GO-TO-MARKET PLAN

- Week 0–2:** test-print the v12 parts, film a 15 s video of a ledge swinging open (the hook for every ad) and take photos on a real wall.
- Launch:** list on Etsy (physical + digital) and on MakerWorld/Printables with a free single rack to build reviews. Post the video to r/HotWheels and to Instagram and TikTok collector groups.
- Meta ads:** \$10–15/day, ages 25–45, interests: Hot Wheels, die-cast, Matchbox, car culture. Test the 5 posters, keep the 2 cheapest per click, retarget site visitors with the “Full strip \$39.99” ad.
- Listing title:** “PITLANE Swing-Out Wall Display for Carded 1:64 Die-cast Cars, fits Hot Wheels® & Matchbox® mainline cards, 7-car 40 cm strip”.

NAME

PITLANE: each ledge is a garage that swings open, like the pit lane at a race. The orange stripe nods to the orange track every collector remembers. Backups: **CardGarage**, **Swing Grid**. Before you register a name, check trademark and domain availability (e.g. USPTO TESS, your local registry, the Etsy shop name).